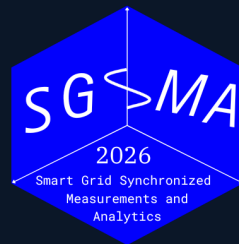


5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*



Physics-Informed PMU Event Detection and Localization

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Date

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Venue

Santiago, Chile



Method overview

Pipeline

1. Convert raw PMU waveforms into physics-aware features.
2. Use a hierarchical ExtraTrees classifier to identify event type.
3. Route each event to a specialized BUS, LINE, or PMU localizer.
4. Produce the challenge CSV output at timestamp resolution.

Reimplementation target

The method can be rebuilt from PMU windows, feature definitions, ExtraTrees models, and typed localization routes.

Submission boundary

The submitted route is SGMS: physics-informed features, hierarchical ExtraTrees, and typed dynamic localizers. RAW-label-tuned and oracle branches are excluded from inference.

Problem and Data

PMU streams, event labels, and the prediction contract

Task, event labels, and outputs

Input

Eight synchronized PMU CSV files: one per observed bus, about 90 min at 30 fps, with voltage/current phasors, frequency, ROCOF, DATA_PRESENT, and hidden or blank Event.

Output

TIMESTAMP, Bus, Predicted_Event, and Predicted_Location.

Label	Name	Meaning
0	normal	no disturbance or major data issue
1	fault	fast, strong voltage disturbance
2	line outage	current redistribution without fault-level voltage collapse
3	generation	frequency/ROCOF and power-proxy shift
4	load	sustained current and power change
5	missing	data loss without physical evidence
6	missing + physical	missing data plus physical event
7	bad data	isolated inconsistent sensor/channel
8	unknown	mixed physical evidence not explained by 1–4

Decision structure: detection, event classification, and localization are coupled, not one flat label.

PMU signals and phasor inputs

Raw columns

Group	Columns
Voltage magnitude	VA_MAG, VB_MAG, VC_MAG
Current magnitude	IA_MAG, IB_MAG, IC_MAG
Voltage angle	VA_ANG, VB_ANG, VC_ANG
Current angle	IA_ANG, IB_ANG, IC_ANG
Dynamics	Freq, ROCOF
Quality/time	DATA_PRESENT, TIMESTAMP

Observed PMUs

Buses 2, 5, 6, 10, 19, 22, 29, and 39.

Measurement model

$$V_{\phi}(t) = |V_{\phi}(t)|e^{j\theta_{\phi}^V(t)}, \quad I_{\phi}(t) = |I_{\phi}(t)|e^{j\theta_{\phi}^I(t)}$$

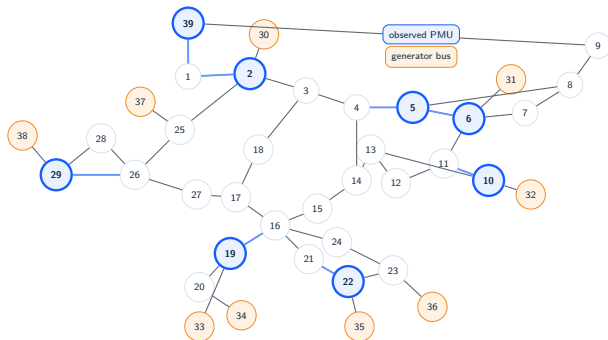
$$\phi \in \{A, B, C\}, \quad \theta_{\text{rad}} = \pi \theta_{\text{deg}} / 180$$

$$\dot{f}(t) = \text{ROCOF}(t)$$

Inference scope

The extractor uses magnitudes, unwrapped angles, frequency, ROCOF, and availability. It does not solve a full AC power-flow at inference time.

IEEE 39-bus system view



Observed PMUs

{2, 5, 6, 10, 19, 22, 29, 39}

Electrical reading

The localizer sees eight sparse PMU views. Events must be inferred from the disturbance pattern across those locations.

Topology used

Electrical distances come from the IEEE 39-bus admittance model, not drawing geometry.

Topology math: from branches to electrical distance

Admittance and impedance

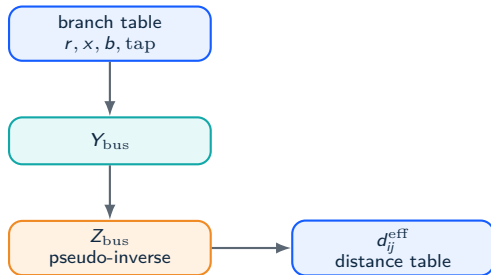
Y_{bus} : network admittance matrix

$$Z_{\text{bus}} = Y_{\text{bus}}^+$$

$$d_{ij}^{\text{eff}} = |Z_{ii} + Z_{jj} - Z_{ij} - Z_{ji}|$$

Meaning

Two buses are close if an injection at one bus is expected to have a similar voltage effect at the other.



Where it appears

Graph-temporal features use $1/d_{ij}^{\text{eff}}$ weights.

Runtime route and end-to-end method

Entrypoint

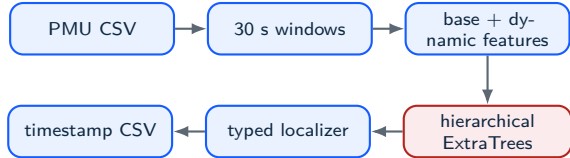
```
python main.py -input-dir <folder>
-output <csv>
```

Active route

```
sgms_extra_trees_windowed, ml_windowed, 30
s windows.
```

Fallback

The bus-agnostic physics runtime is only a missing-bundle fallback.



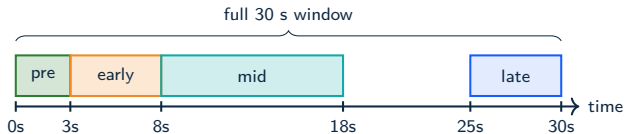
Runtime shape

Each window becomes one physics-aware feature row. The output is expanded back to timestamp-bus rows for the official CSV.

Windowing and Baseline

30-second windows and segment statistics

30-second window and segment design



Segments

$$S_{\text{pre}} = [0, 3), \quad S_{\text{early}} = [3, 8)$$

$$S_{\text{mid}} = [8, 18), \quad S_{\text{late}} = [25, 30)$$

$$S_{\text{full}} = [0, 30)$$

Reason

The window is long enough to capture onset, transient, and settling, but short enough to produce one stable event decision.

Local baseline, robust scale, and subtraction

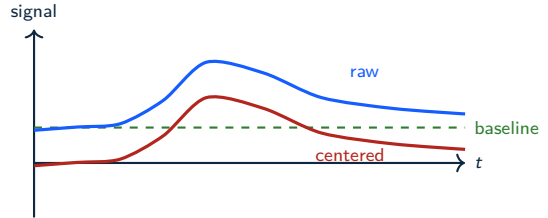
Per PMU and signal

$$b_{p,s} = \text{median}_{t \in S_{\text{pre}}} x_{p,s}(t)$$

$$\sigma_{p,s}^{\text{rob}} = 1.4826 \text{ median}_{t \in S_{\text{pre}}} |x_{p,s}(t) - b_{p,s}| + \epsilon$$

$$x'_{p,s}(t) = x_{p,s}(t) - b_{p,s}$$

$$z_{p,s}(t) = |x'_{p,s}(t)| / \sigma_{p,s}^{\text{rob}}$$



Scale fallback

If MAD is too small, the code falls back to standard deviation and then to a safe value.

Base Feature Engineering

Per-window statistical feature families

Feature schema and base families

Canonical feature key

<PMU/BUS>__<SIGNAL>__<SEGMENT>__<STAT>

Bus2__VA_MAG__early__span

Bus10__Freq__late__mean

GLOBAL__single_signal_score_max

Meaning

The name stores source, physical signal, temporal segment, and statistic.

Family	Purpose
Segment statistics	magnitude, spread, persistence
Temporal deltas	onset and post-event shift
Derivatives	fast transients and ramps
Robust z-score	scale-normalized abnormality
Sustained anomaly	duration above threshold
Data quality	NaNs, gaps, missing runs
Bad-data score	isolated channel faults
Phase spread	three-phase coherence
Power proxy	rough P/Q movement
Global aggregation	system-wide maxima and counts

Temporal deltas and derivative features

Definitions

$$\Delta_{\text{early}} = \mu_{\text{early}} - \mu_{\text{pre}}$$

$$\Delta_{\text{mid}} = \mu_{\text{mid}} - \mu_{\text{pre}}$$

$$\Delta_{\text{late}} = \mu_{\text{late}} - \mu_{\text{pre}}$$

Interpretation

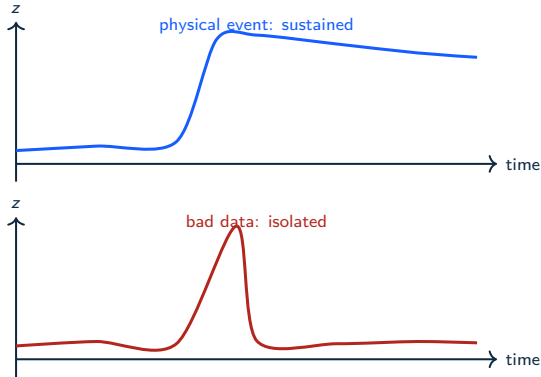
- early delta: onset shock.
- mid delta: transient settling.
- late delta: persistent operating-point change.

Derivative features

$$d_x(t) = \frac{x(t) - x(t - \Delta t)}{\Delta t}, \quad \max_t |d_x(t)|, \quad \sqrt{T^{-1} \sum_t d_x(t)^2}$$

Large voltage/current derivatives flag abrupt transients used by the fault and line-outage guards.

Robust z-score separates two failure modes



Peak and persistence

`max_abs_robust_z` catches the spike. Sustained features say whether the abnormal value stays high for several samples.

Classifier use

`sustained_abs_robust_z_5` and `sustained_abs_robust_z_15` separate short bursts, persistent physical deviations, and isolated bad-data spikes.

Data quality, bad data, and phase consistency

Missing-data features

- `n_samples`, `duration_s`, `dt_s`
- `max_timestamp_gap_s`, `max_timestamp_gap_ratio`
- `data_present_fraction`, `data_present_min`
- `missing_run_max_samples/s`
- `nan_fraction_max/mean`

Missing route

Missing only → Event 5. Missing plus physical evidence
→ Event 6.

Bad-data score

$$\frac{\text{top anomaly}}{\text{second anomaly}} \geq 3 \Rightarrow \text{bad data}$$

Phase consistency

Phase-to-phase spread and angle coherence
separate grid events from corrupted single
channels.

Power proxy, global aggregation, and base count

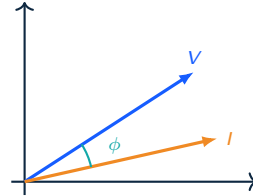
Per-unit magnitudes

$$V_{pu,\phi}(t) = \frac{|V_{\phi}(t)|}{\max(\text{median}_{pre}|V_{\phi}|, \epsilon)}$$

$$I_{pu,\phi}(t) = \frac{|I_{\phi}(t)|}{\max(\text{median}_{pre}|I_{\phi}|, \epsilon)}$$

Proxy, not calibrated power

The challenge data does not require a full power-flow solve. Sign depends on the PMU current convention, but the proxy gives the classifier a stable active/reactive signature.



$$\begin{aligned} P_{\text{proxy}} &= V_{pu} I_{pu} \cos \phi \\ Q_{\text{proxy}} &= V_{pu} I_{pu} \sin \phi \end{aligned}$$

Global summaries and count

Stores cross-PMU mean, max, min, std. Base statistical total: **39,744** features.

Dynamic Feature Engineering

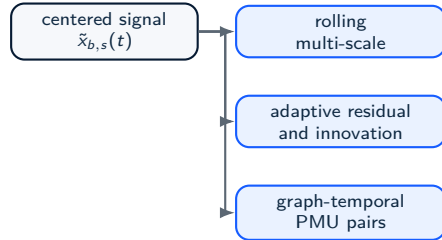
Rolling, adaptive-residual, and graph-temporal evidence

Dynamic features: signals and three views

Signals

{VA_MAG, IA_MAG, VA_ANG,
IA_ANG, Freq, ROCOF}

Full magnitude/angle families are used where feasible; the reduced set controls feature count for residual and graph modules.



Purpose

These modules add time-scale, residual, and spatial-propagation evidence after the base statistics.

Rolling multi-scale features and equations

Rolling windows

$$w \in \{5, 15, 30, 60, 120\}$$

Features

- mean_abs_max
- std_max, std_mean
- slope_max_abs
- energy
- time_to_abs_peak

Equations

$$\mu_w(t) = w^{-1} \sum_{\tau \in W_w(t)} \tilde{x}(\tau)$$

$$\sigma_w(t) = \text{std}_{\tau \in W_w(t)} \tilde{x}(\tau)$$

$$E_x = T^{-1} \sum_t |\tilde{x}(t)|^2$$

Interpretation

Short windows capture abrupt switching/fault onset.
Long windows capture sustained load/generation shifts.
Time-to-peak captures delay.

RLS level and linear-trend residuals

Model

$$y_t = \theta_t + e_t$$

$$K_t = \frac{P_{t-1}}{\lambda + P_{t-1}}$$

$$\theta_t = \theta_{t-1} + K_t(y_t - \theta_{t-1})$$

$$P_t = \frac{P_{t-1} - K_t P_{t-1}}{\lambda}$$

Forgetting factors

$$\lambda \in \{0.92, 0.985\}$$

$$\phi_t = [1, t - t_0]^T, \quad \hat{y}_t = \phi_t^T \theta_{t-1}$$

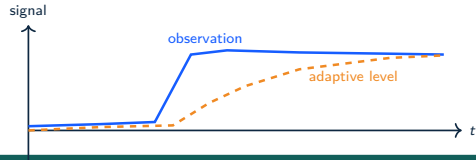
Features produced

$$r_t = |y_t - \theta_{t-1}|$$

$$\Delta\theta_t = |\theta_t - \theta_{t-1}|$$

For each sequence:

max, mean, p95,
median_{late} - median_{pre}



Graph-temporal features and decision form

For signal s and observed buses $\mathcal{B}$

$$\text{spread}_s(t) = \max_{b \in \mathcal{B}} \tilde{x}_{b,s}(t) - \min_{b \in \mathcal{B}} \tilde{x}_{b,s}(t)$$

$$\Delta_{ab,s}^{\max} = \max_t |\tilde{x}_{a,s}(t) - \tilde{x}_{b,s}(t)|$$

$$\rho_{ab,s} = \text{corr}(\tilde{x}_{a,s}, \tilde{x}_{b,s})$$

$$\Delta t_{ab,s}^{\text{peak}} = |t_{a,s}^{\text{peak}} - t_{b,s}^{\text{peak}}|$$

Electrical weighting

$$w_{ab} = \frac{1}{\max(d_{ab}^{\text{eff}}, 10^{-5})}$$

$$G_s^{\max} = \max_{a < b} w_{ab} \Delta_{ab,s}^{\max}$$

Decision form

$$x_{\text{loc}} = [x_{\text{base}}, x_{\text{dyn}}, G]$$

$$\hat{\ell} = \arg \max_{\ell} p_{\text{ET}}(\ell \mid x_{\text{loc}})$$

Graph equations create columns; ExtraTrees learns the thresholds and pair combinations.

Hierarchical ML Method

ExtraTrees detection, classification, and event routing

The ML model in one slide

What model

ExtraTrees (Extremely Randomized Trees), from `scikit-learn`.
Each model = `SimpleImputer(median)` + `ExtraTreesClassifier`.
Not a neural network: 0 trainable NN parameters, CPU-only, deterministic.

Four models, same feature row X , different label y

1. Physical event classifier → events 0,1,2,3,4,8
2. Bad-data detector → event 7
3. Missing detector → events 5 vs 6
4. Typed localizers → BUS / LINE / PMU

Why ExtraTrees here

- Handles 45,162 features with no scaling.
- Tolerates NaN via median imputer (real PMU gaps).
- Works with few labeled windows.
- Interpretable: feature thresholds + vote.
- Tiny bundle, fast CPU inference.

Forest vote

$$\hat{y} = \arg \max_c \frac{1}{T} \sum_{t=1}^T p_t(c | X)$$

Which feature explains which event

Event	Discriminating feature (physics)	Localizer
1 fault	Voltage span V_{span} + voltage derivative D_V + robust z (fast deep V sag)	BUS (spatial severity)
2 line outage	Current span I_{span} + current-angle change + graph pairwise diff, with V_{guard}	LINE (graph candidates)
3 generation	Freq shift + ROCOF span + P/Q proxy (sustained frequency/power move)	BUS
4 load	Sustained current shift + active-power proxy + rolling energy	BUS
5 missing	data-present fraction <1 , NaN max, timestamp-gap ratio	PMU
6 missing+phys	missing gate <i>AND</i> physical evidence present	physical route
7 bad data	one channel huge robust z , others coherent (isolated spike)	PMU (the sensor)
8 unknown	physical evidence + low confidence for events 1–4	BUS

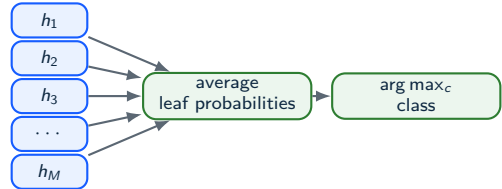
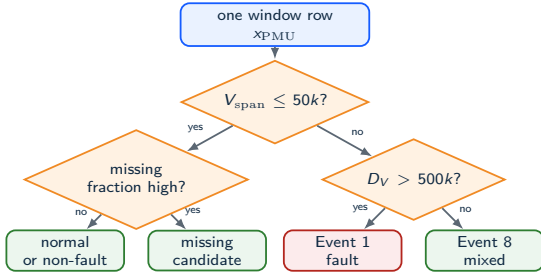
Four feature families (physics meaning)

Base statistics + derivatives → *abruptness*; robust z → *anomaly vs noise*; RLS/Kalman innovation → *break vs ramp*; graph-temporal → *spatial propagation*.

One-line answer if asked

“V-channels detect faults, I-channels detect outages, Freq/ROCOF/PQ detect gen/load, data-quality detects 5/6/7, and graph-temporal features localize.”

ExtraTrees: one tree and the forest vote



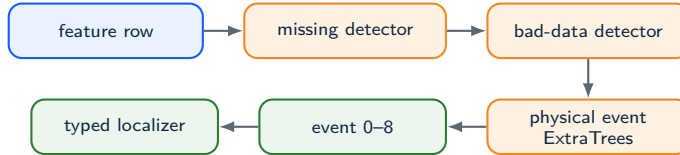
Forest probability

$$p_m(c | x) = \frac{n_{\text{leaf},c}}{n_{\text{leaf}}}, \quad \hat{p}(c | x) = \frac{1}{M} \sum_m p_m(c | x)$$

$$\hat{c} = \arg \max_c \hat{p}(c | x)$$

One tree follows a threshold path; ExtraTrees averages many randomized trees, so the submitted decision is a probability vote, not one manual rule.

Hierarchical event routing



Decision order

Missing and bad-data evidence is routed before the physical classifier so communication failures and sensor artifacts do not contaminate physical events.

Physics rule guards and tree decision

Main guards

$V_{\text{span}} \geq 50,000 \vee D_V \geq 500,000 \Rightarrow \text{fault candidate}$

$I_{\text{span}} \geq 500, \quad D_I \geq 3,000, \quad V_{\text{span}} \leq 50,000$

$\Rightarrow \text{line outage candidate}$

$\text{single_signal_score} \geq 200,000, \quad \text{ratio} \geq 3 \Rightarrow \text{bad data}$

Tree role

ExtraTrees estimates class probabilities from the full engineered vector. Rule guards override only high-risk confusion points: missing, bad data, fault, and line outage.

Output

$$\hat{e} = \arg \max_e p(e | x), \quad \hat{\ell} = \arg \max_{\ell} p(\ell | x, \hat{e})$$

Event-by-Event Interpretation

How each event type is recognized

Event 1: fault

Main evidence

- Voltage magnitude span.
- Voltage derivative.
- Robust z-score in voltage channels.
- Early segment changes.
- Cross-PMU severity pattern.

Manual recipe

1. Build baseline from pre-event voltage.
2. Detect large voltage sag/transient.
3. Reject missing-only and bad-data-only cases.
4. Localize to BUS using spatial severity.

Event 2: line outage

Main evidence

- Current magnitude span.
- Current derivative.
- Current-angle changes.
- Voltage span not as extreme as a fault.
- Graph-temporal pairwise PMU differences.

Manual recipe

1. Look for current redistribution.
2. Guard against fault-like voltage collapse.
3. Use graph features to score candidate lines.
4. Localize to LINE.

Event 7: bad data

Main evidence

- One signal has extreme robust z-score.
- Top anomaly is much larger than second anomaly.
- Other PMUs/channels remain coherent.
- Missing features do not dominate.

Localizer

Event 7 localizes to PMU, not BUS or LINE. The object that failed is the measurement device/channel source.

Event 8: unknown physical proxy

Main evidence

Event 8 captures cases with physical evidence that does not cleanly fit fault, line outage, generation change, or load change.

Useful features

- Power proxy.
- Rolling dynamics.
- Adaptive residual and innovation features.
- Graph-temporal pattern.

Routing

The final runtime routes Event 8 to a BUS localizer because it represents a physical-grid signature rather than a PMU-only failure.

Localization

Typed localizers and the hybrid bus estimate

Typed origin localization

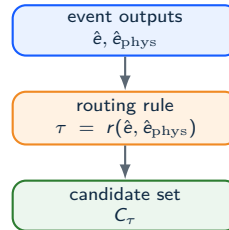
Route before predicting origin

Predicted event	Candidate origin set
0 normal	NONE
1 fault	BUS labels
2 line outage	LINE labels
3 generation change	BUS labels
4 load change	BUS labels
5 missing data	PMU labels
6 missing + physical	physical route
7 bad data	PMU labels
8 mixed physical proxy	BUS labels

For event 6: $\hat{e}_{\text{phys}} = 2 \Rightarrow \text{LINE}$; otherwise BUS.

Implemented decision

The model does not rank every possible label together. It first chooses a type, then runs the localizer trained for that type.



Final localizer models

Model lookup used at inference

First choice	Fallback
BUS:event1, BUS:event3, BUS:event4, BUS:event8	BUS
LINE:event2	LINE
PMU	PMU

Model family

Each entry is a median-imputed ExtraTrees pipeline trained on the localizer feature vector. Event-specific models are used when the physical event prediction selects one.

Final Top-1

use_hybrid_top1 = false. The final submitted Top-1 location uses the validated typed ML localizers, not the hybrid/oracle branch.

Localizer evidence and inference equation

Evidence used

- local PMU amplitude and robust z-score;
- derivative and residual/innovation peak timing;
- graph-temporal pair differences and electrical-distance weights;
- event-specific candidate space: BUS, LINE, or PMU.

Inference

$$p_{\tau}(\ell \mid x) = \frac{1}{T} \sum_{m=1}^T \mathbf{1}\{h_m(x) = \ell\}$$

$$\hat{\ell} = \arg \max_{\ell \in \mathcal{L}_{\tau}} p_{\tau}(\ell \mid x)$$

Interpretation

The event route selects the localizer type τ , then the ExtraTrees model ranks candidates only inside $\mathcal{L}_{\tau}$.

Training, Validation, and Results

Evaluation protocol and reported metrics

Training data and split

Training sources

- 5,000 synthetic SGSMA scenario windows.
- 3,500 train windows and 1,500 stratified holdout windows.
- Base statistical and dynamic tables generated by the repo feature extractors.
- Random state: 20260503.

Validation

- 30% stratified SIM holdout.
- RAW0001 transfer validation.
- Metrics: detector, classifier, macro-F1, localization.

Transfer principle

RAW labels were used for validation and analysis, not to train the final production-feasible submission path.

Final metrics

SIM holdout

Metric	Score
Detector accuracy	0.9693
Classifier accuracy	0.9673
Classifier macro-F1	0.9598
Localizer Top-1	0.8524

RAW0001

Metric	Score
Detector accuracy	1.0000
Classifier accuracy	1.0000
Classifier macro-F1	1.0000
Localizer Top-1	0.6667

Interpretation

Detection and classification transferred perfectly on RAW0001; localization remained the hard part.

Variant selection evidence

Selected variant

base statistical + rolling + adaptive residual + graph-temporal

Selection evidence

- SIM detector accuracy: 0.9693.
- SIM classifier macro-F1: 0.9598.
- RAW0001 localization Top-1: 0.6667.
- No oracle RAW label use.
- No large NN/GNN training-data requirement.

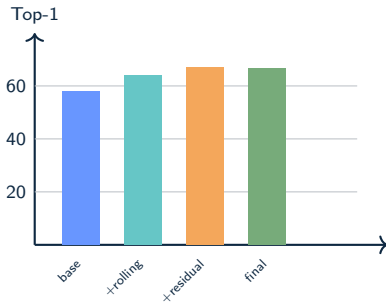
Tradeoff

Some experimental methods had higher RAW0001 numbers, but were not valid for unseen data because they used RAW true labels or oracle assumptions.

Ablation and Reimplementation

What each component contributes

Ablation evidence



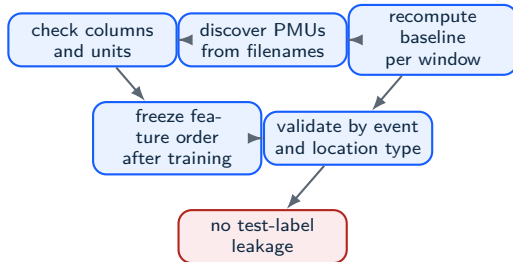
Selection evidence

The selected variant tied the best deployable RAW0001 localization result while keeping SIM validation strong.

Rejected variants

RAW-label-trained, oracle, or leakage-dependent variants were not deployable on a new hidden RAW dataset.

Adapting to a new RAW dataset



Check first

Sampling rate, timestamp continuity, PMU list, channel names, angle units, and missing-data coding.

Keep stable

If the schema matches RAW0001, run the frozen model:
`main.py -input-dir <RAW0002> -output <predictions.csv>.`

No hidden-label tuning

Retrain or recalibrate only with training labels. Hidden-test labels must not affect feature selection, thresholds, or localizer ranking.

Reimplementation Sequence

Rebuilding the method from zero

How the ExtraTrees forest is trained

Growing one tree (greedy split, random cuts)

Training rows are labeled windows (x, y) : $x \in \mathbb{R}^{\approx 45k}$ features, y the event class. At every node:

1. draw K random features (not all 45k);
2. pick *one random* threshold for each;
3. keep the cut with the best impurity drop;
4. recurse until leaves are pure or `min_samples_leaf`.

The “Extra” part

Thresholds are *random*, not optimized per feature, and each tree sees the *full* set (no bootstrap). This makes trees fast and diverse, which lowers variance.

Split score: Gini impurity

$$G = 1 - \sum_c p_c^2, \quad \Delta G = G_{\text{node}} - \sum_{\text{child}} \frac{n}{N} G_{\text{child}}$$

Keep the random cut with the largest ΔG .

From tree to forest

Grow M independent trees with different random features and cuts. Each leaf stores class frequencies; the forest averages the leaf probabilities. Tuned controls are `n_estimators`, `max_features`, and `min_samples_leaf`: CPU-only, no scaling, NaN via median imputer.

Why a learned forest, not hand-written if/else

Why if/else on our own features fails

- $\approx 45,000$ features, 9 event classes, 3 localizer types: hand-tuning thresholds is infeasible and brittle.
- One cutoff per feature misses *interactions*: high robust- z *and* sustained *and* low missing $\Rightarrow$ fault; high z alone $\Rightarrow$ bad data.
- Hand-picked cutoffs overfit RAW0001 noise and scale, and break on a hidden RAW with different PMUs.

What the forest does instead

- Learns thresholds *and* feature combinations from labeled windows; finds the few discriminative ones among 45k.
- Emits calibrated probabilities $\rightarrow$ feed the physics gates (bad-data, missing) and the confidence checks.
- Tolerates correlated/redundant features and NaN, needs no scaling, and is deterministic, tiny, and fast.

Key idea: a forest averages learned randomized threshold cascades, finding feature combinations that would be brittle to tune by hand.

ExtraTrees route versus a neural network

Aspect	Submitted ExtraTrees route	NN/GNN alternative
Input	One engineered row per 30 s window: electrical statistics, residuals, graph features.	Raw or lightly processed PMU tensor: buses $\times$ time $\times$ channels.
Learning target	Trees learn threshold combinations over features. Rules enforce missing/bad-data and physics guards.	Weights learn representation and classifier jointly; GNN adds message passing over topology.
Data need	Works with limited labeled windows because features encode much of the physics.	Needs more labeled scenarios, augmentation, and stronger regularization to avoid RAW overfit.
Decision trace	Leaf probabilities, feature thresholds, event gates, typed localizer route.	Usually less transparent; explanations require saliency, attention, or post-hoc tools.
Deployment	Joblib bundle + JSON config + deterministic CPU inference.	Model checkpoint + scalars + architecture code; more sensitive to shape/schema drift.
Best future use	Strong baseline and interpretable competition submission.	Professional next step when larger labeled RAW/SIM datasets and topology metadata are available.

Practical conclusion

The final method chose reliability and auditability over end-to-end representation learning.

Closing

Submitted method

The final submission transforms raw PMU windows into physically meaningful feature vectors, then uses hierarchical ExtraTrees models and typed dynamic localizers to produce event and location predictions.

Feature engine

Base statistical features + rolling dynamics + adaptive residuals + graph temporal features.

Classifier

Physical event model + missing detector + bad-data detector + rules.

Localizer

Specialized BUS, LINE, and PMU ExtraTrees localizers.

Selected features: base statistical + rolling + residual/innovation + graph temporal

Conclusion

Core method	Why it worked	Localization
Raw PMU windows are converted into physically meaningful feature rows. Hierarchical ExtraTrees models classify the event type.	Robust statistics capture abnormal behavior. Residual, innovation, and graph-temporal features add timing and network evidence.	Typed localizers keep BUS, LINE, and PMU targets separate. Each model ranks valid candidates inside its own target space.

Final takeaway

The submitted system is reproducible because every output comes from a feature row, a routing rule, or an ExtraTrees probability vote.